

Trainee Program

DTPP — Deshmukh Technologies Trainee Program

A 90-day full-stack software development and professional development program that prepares trainees to build, test, debug, explain, document, and maintain production software.

PRIMARY OUTCOME

Junior Full-Stack Developer

BACKEND TRACKS

Java / Spring Boot or Python / FastAPI

DELIVERY

Git + GitHub + Docker + CI/CD + AWS

FRONTEND

React + TypeScript

DATABASE

MySQL / PostgreSQL

DURATION

90 Days · 12 Phases

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I. Program Purpose

The Deshmukh Technologies Trainee Program (DTTP) is a structured professional development program designed to transform freshers, interns, graduates, and entry-level candidates into capable software engineers.

The program combines technical education, practical coding, full-stack development, project-based learning, engineering practices, mentorship, code review, testing, debugging, security, DevOps, professional development, and continuous assessment.

The program does not define success as completion of 90 days. Success means the trainee can demonstrate the ability to build, test, debug, explain, document, and maintain software.

2. Program Vision

Build skilled, responsible and production-ready software professionals who can understand business requirements, develop complete applications, solve problems independently and contribute effectively to real Deshmukh Technologies projects.

3. Program Mission

1. Establish strong programming fundamentals.
 2. Develop modern full-stack development skills.
 3. Teach professional software engineering practices.
 4. Provide practical project experience.
 5. Develop independent problem-solving ability.
 6. Build communication and teamwork skills.
 7. Introduce testing, security and deployment.
 8. Measure capability continuously.
 9. Provide a structured path from trainee to developer.
 10. Create a repeatable internal talent-development system for Deshmukh Technologies.
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4. Three-Dimensional Development Model

Every trainee is evaluated across three dimensions. A trainee must develop all three.

A. Technical Capability

- Programming
- Frontend
- Backend
- Database
- APIs
- Testing
- Security
- Cloud

B. Engineering Capability

- Git
- Architecture
- Clean code
- Debugging
- Code review
- CI/CD
- Docker
- Documentation

C. Professional Capability

- Communication
 - Discipline
 - Teamwork
 - Ownership
 - Time management
 - Accountability
 - Problem solving
-

5. Technology Architecture

COMMON FOUNDATION

FRONTEND

```
PATH
HTML
↓
CSS
↓
JavaScript
↓
TypeScript
↓
React
```

```
PATH
Programming
↓
OOP
↓
Data Structures
↓
Algorithms
↓
Web Fundamentals
↓
SQL
↓
Git / GitHub
↓
REST APIs
```

BACKEND — TRACK A

```
JAVA
Java
↓
Spring Boot
↓
Spring Data JPA
↓
Hibernate
```

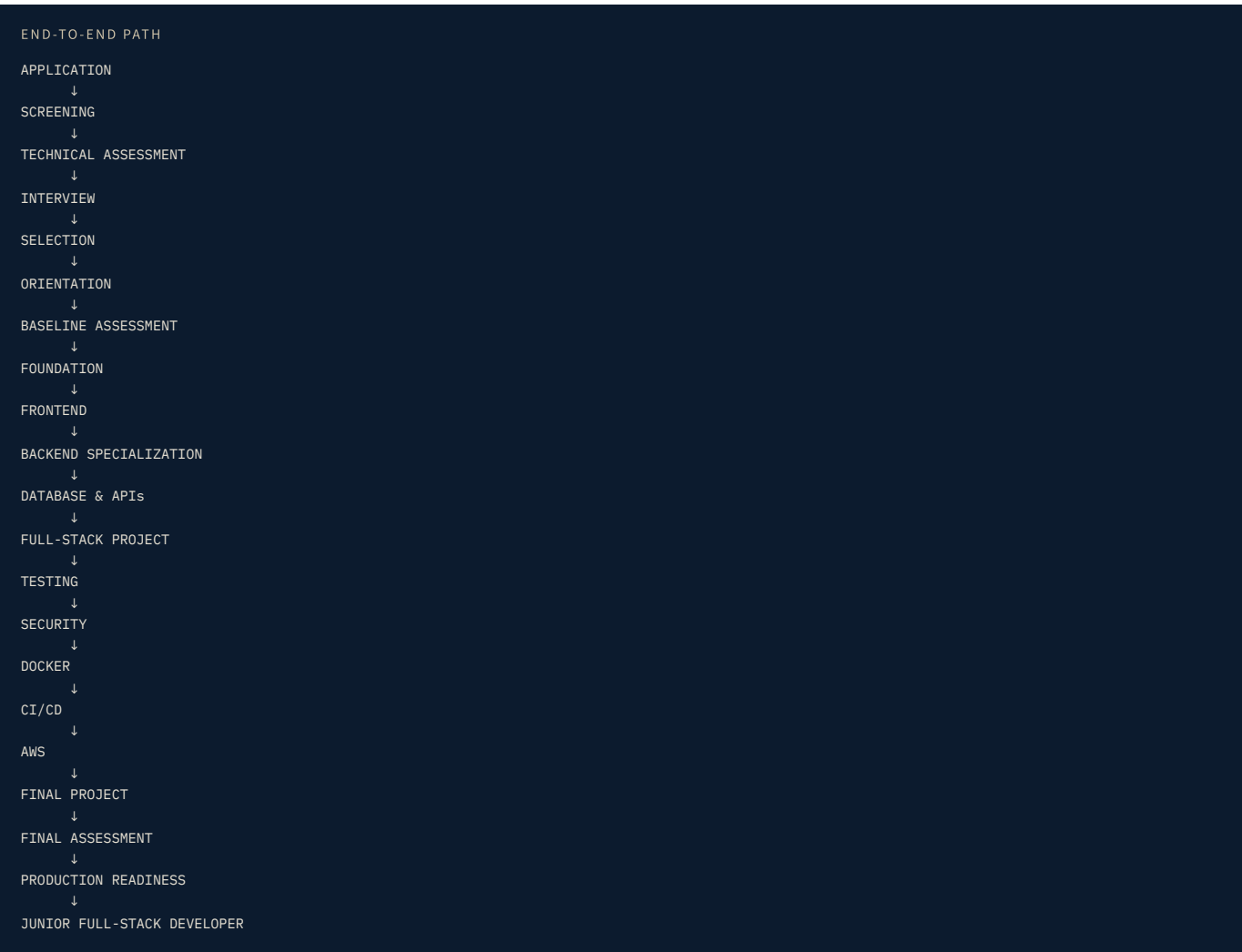
BACKEND — TRACK B

```
PYTHON
Python
↓
FastAPI
↓
SQLAlchemy
```

ENGINEERING

```
DELIVERY STACK
Testing → Security → Docker → CI/CD → AWS → Monitoring
```

6. Trainee Journey



7. 90-Day Master Roadmap

Phase	Days	Outcome
1	1–5	Professional and engineering foundation
2	6–15	Programming and problem solving
3	16–25	Web and frontend foundation
4	26–35	React + TypeScript
5	36–45	Java / Spring Boot or Python / FastAPI
6	46–55	Database + REST API
7	56–65	Full-stack project
8	66–72	Authentication + security
9	73–78	Testing + debugging
10	79–82	Docker
11	83–86	CI/CD + AWS
12	87–90	Final project + assessment

8. Training Methodology

Recommended allocation of trainee time:

Activity	Percentage
Theory	20%
Practical	30%
Project	40%
Review & Assessment	10%

CORE PRINCIPLE

The trainee should spend more time building software than listening to someone explain software.

9. Daily Development Cycle

HABIT LOOP

LEARN → UNDERSTAND → PRACTICE → IMPLEMENT → TEST → DEBUG → COMMIT → PULL REQUEST → CODE REVIEW → IMPROVE

This should become the trainee's normal development habit.

10. Phase 1 — Orientation

Days 1–5. Orientation establishes company context and professional standards before technical training begins.

COMPANY

- Deshmukh Technologies
- Vision and mission
- Products and services
- Projects and teams
- Development methodology
- Customer expectations

PROFESSIONAL STANDARDS

- Attendance and working hours
- Communication and meetings
- Workplace conduct
- Confidentiality
- Information security
- Responsibility

11. Baseline Assessment

Before technical training begins, assess the trainee and record a **Baseline Skill Profile**.

PROGRAMMING

- Logic
- Variables
- Conditions
- Loops
- Functions

TECHNICAL

- Java
- Python
- JavaScript
- SQL
- Git

PROBLEM SOLVING

- Logical reasoning
- Coding problem

COMMUNICATION

- Self-introduction
- Technical explanation
- Problem explanation

12. Git & GitHub

Git is mandatory. Trainees must learn repository, clone, commit, branch, push, pull, merge, rebase, stash, revert, pull request, code review, and conflict resolution.

STANDARD WORKFLOW

DELIVERY WORKFLOW

TASK → BRANCH → CODE → TEST → COMMIT → PUSH → PULL REQUEST → REVIEW → FIX → APPROVE → MERGE

13. Phase 2 — Programming

Days 6–15.

COMMON PROGRAMMING

[Variables](#)[Data types](#)[Operators](#)[Conditions](#)[Loops](#)[Functions](#)[Collections](#)[Strings](#)[Exceptions](#)[OOP](#)[Data structures](#)[Algorithms](#)

BEGINNER EXERCISES

- Reverse string
- Palindrome
- Fibonacci
- Factorial
- Prime
- Largest element

INTERMEDIATE EXERCISES

- Two Sum
- Anagram
- Missing number
- Duplicate detection
- Frequency counting
- Sorting, searching, LinkedList

14. Java Track

JAVA FUNDAMENTALS

- OOP, classes, interfaces
- Inheritance, polymorphism, encapsulation
- Exceptions
- Collections
- Generics

MODERN JAVA

- Lambda
- Streams
- Optional
- Functional interfaces
- Date/time API

15. Spring Boot

ARCHITECTURE

Controller → Service → Repository → JPA / Hibernate → Database

Spring Boot

REST

Dependency Injection

DTO

Entity

Validation

Exception handling

Logging

JPA

Hibernate

Spring Security

JWT

Testing

16. Python Track

Variables

Lists

Tuples

Sets

Dictionaries

Functions

Modules

Packages

Exceptions

File handling

OOP

Type hints

Generators

Decorators

Async

17. FastAPI

ARCHITECTURE

Router → Service → Repository → SQLAlchemy → Database

Routes

Request models

Response models

Pydantic

Validation

Dependency injection

Middleware

Exceptions

Authentication

Async

API documentation

18. Frontend — React + TypeScript

Trainees learn components, props, state, hooks, forms, validation, routing, protected routes, API integration, error handling, loading states, and responsive design.

REACT ARCHITECTURE

```
SRC/  
  
src/  
├ components/  
├ pages/  
├ layouts/  
├ services/  
├ hooks/  
├ types/  
├ utils/  
├ routes/  
└ assets/
```

19. Database

Trainees learn SQL, tables, keys, constraints, relationships, joins, indexes, transactions, normalization, and query optimization.

ORM: Java uses JPA / Hibernate. Python uses SQLAlchemy.

ORM knowledge must not replace SQL knowledge.

20. REST API

Both backend tracks implement the same business API so Java and Python trainees learn the same API design principles.

```
EMPLOYEE MANAGEMENT API  
  
POST  /api/auth/login  
  
GET   /api/employees  
POST  /api/employees  
GET   /api/employees/{id}  
PUT   /api/employees/{id}  
DELETE /api/employees/{id}  
  
GET   /api/departments  
POST  /api/departments  
  
GET   /api/attendance  
POST  /api/attendance  
  
GET   /api/leaves  
POST  /api/leaves  
PUT   /api/leaves/{id}/approve
```

21. Major Training Project

Employee Management System

Roles: ADMIN · HR · EMPLOYEE

Authentication	Dashboard	Employee Management	Department Management	Attendance	Leave	Reports	Profile
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22. Full-Stack Architecture

JAVA IMPLEMENTATION

React → REST → Spring Boot → Service → JPA / Hibernate → MySQL / PostgreSQL

PYTHON IMPLEMENTATION

React → REST → FastAPI → Service → SQLAlchemy → MySQL / PostgreSQL

Both implementations must satisfy the same business requirements.

23. Authentication & Security

- Authentication and authorization
- Password hashing, JWT, roles, and permissions
- CORS, input validation, SQL injection awareness, XSS fundamentals
- Secrets management, environment variables, HTTPS fundamentals, least privilege

Never commit passwords, API keys, tokens, private keys, or production credentials to GitHub.

24. Testing

Backend Java: JUnit, Mockito, Spring Boot tests Python: Pytest, FastAPI testing, mocking	Frontend • Component testing • Form testing • API-state testing • Error-state testing	API • Postman • Positive and negative tests • Authentication and authorization • Edge cases
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25. Debugging

METHOD

BUG → REPRODUCE → READ ERROR → CHECK LOGS → CHECK NETWORK → CHECK API → CHECK BACKEND → CHECK DATABASE → ROOT CAUSE → FIX → RETEST → REGRESSION TEST

The trainee should learn to determine which layer is responsible rather than randomly changing code.

26. Docker

CONTAINERS

Frontend Container → Backend Container → Database Container

Learn Dockerfile, image, container, network, volume, environment variables, and Docker Compose.

27. CI/CD

EXAMPLE PIPELINE

Developer → GitHub → Build → Lint → Unit Tests → Integration Tests → Docker Build → Artifact → Deployment → Health Check

28. AWS

Introduce IAM, EC2, S3, RDS, CloudFront, Route 53, Security Groups, and CloudWatch fundamentals. The trainee should understand the purpose of each service before using it.

29. Project Management

Every trainee project should use a professional work breakdown.

Level	Example
Epic	Employee Management
Features	Authentication, Employees, Departments, Attendance, Leave, Reports
User Story	As an administrator, I want to create an employee so that the employee can be managed in the system.
Task	Create employee POST API.
Subtasks	Entity, DTO, Repository, Service, Controller, Validation, Test

This teaches trainees how professional teams convert requirements into engineering work.

30. Definition of Done

A feature is complete only when all of the following are true:

☐ Requirement understood☐ Acceptance criteria satisfied☐ UI implemented☐ API implemented☐ Database completed☐ Validation added☐ Security checked☐ Tests written☐ Tests passed☐ Error cases checked☐ Git commit created☐ Pull Request created☐ Code review completed☐ Review comments resolved☐ Documentation updated☐ Mentor approved

31. Mentor System

Each trainee should have **1 Primary Mentor** and, where practical, **1 Technical Reviewer**.

MENTOR KPIS

- Trainee progress
- Code-review quality
- Feedback quality
- Skill-gap identification
- Project completion
- Production-readiness improvement

This prevents mentorship from becoming informal or inconsistent.

32. Daily Report

Every trainee submits the following:

Field	Question
Daily Objective	What was planned?
Completed	What was actually completed?
Learning	What was learned?
Problems	What went wrong?
Investigation	What was checked?
Resolution	How was it fixed?
Git Activity	Commits / PRs
Next Plan	Tomorrow's objectives

33. Weekly Review

The mentor reviews technical progress, assignment completion, project progress, code quality, problem solving, communication, discipline, Git activity, blockers, and skill gaps.

34. Competency Model

Level	Definition
L0	Not exposed
L1	Understands
L2	Performs with guidance
L3	Performs independently
L4	Can review / explain
L5	Can own

Graduation target: most core skills should reach **L3**. Advanced areas such as architecture, CI/CD, and AWS may initially target **L2**.

35. Performance Status

 Green On track.	 Amber Needs improvement.	 Red At risk.
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Amber and Red trainees receive an **Individual Improvement Plan**.

36. Weekly Score

Area	Weight
Technical Learning	15
Practical Work	20
Problem Solving	15
Project Contribution	20
Code Quality	10
Testing	5
Communication	5
Discipline	5
Documentation	5
Total	100

37. Final Assessment

Category	Weight
Programming	15%
Frontend	10%
Backend	15%
Database / API	10%
Final Project	20%
Problem Solving	10%
Testing / Debugging	5%
Git / CI/CD	5%
Documentation	5%
Communication / Professionalism	5%
Total	100%

38. Graduation Levels

85–100 • Developer Ready Eligible for Junior Full-Stack Developer responsibilities.	70–84 • Junior Developer — Supervised Can contribute with closer mentoring.
55–69 • Extended Training Requires additional structured training.	Below 55 • Management Review Formal decision required.

39. Production Readiness Gate

Before graduation, the trainee must demonstrate:

☐ Requirement understanding	☐ Debugging
☐ System design	☐ Git
☐ React development	☐ Pull Requests
☐ Backend development	☐ Code review
☐ SQL	☐ Docker
☐ REST APIs	☐ CI/CD understanding
☐ Authentication	☐ Deployment understanding
☐ Authorization	☐ Documentation
☐ Testing	☐ Technical communication

40. Career Progression



Promotion must be based on **demonstrated capability**, not simply completion of the 90-day period.

41. DTTP Management System

The long-term goal is an internal Trainee Management System.

ORGANIZATION DASHBOARD

- Total trainees
- Active trainees
- Completed
- Extended
- At-risk

INDIVIDUAL RECORD

- Attendance and daily reports
- Assignments and technical scores
- Git activity, PRs, and code reviews
- Project progress and skill matrix
- Mentor rating and final result

COMPLETE TRAINEE RECORD

PROFILE → BASELINE → ATTENDANCE → DAILY REPORTS → ASSIGNMENTS → WEEKLY REVIEWS → SKILL MATRIX → PROJECT → CODE REVIEWS → FINAL ASSESSMENT → PRODUCTION



42. Final DTTP Standard

The program should not ask: “How many technologies did the trainee complete?”

What can this trainee independently build, test, debug, explain, document and maintain?

TRANSFORMATION

STUDENT



TRAINEE



PRACTITIONER



PROJECT CONTRIBUTOR



INDEPENDENT TRAINEE



PRODUCTION-READY



JUNIOR FULL-STACK DEVELOPER

DESHMUKH TECHNOLOGIES

DTTP — Deshmukh Technologies Trainee Program

Learn. Build. Solve. Review. Deploy. Grow.

A disciplined, technically capable and production-ready
Junior Full-Stack Developer.